SIL10: APPLICATION
Details available at zulfr.com

Issue:

Reliability models for Safety Instrumented Systems (SIS) historically rely on stationary parameters. Constant failure rates, fixed diagnostic coverage, and invariant common-cause factors establish a baseline for safety integrity. These models provide a necessary foundation for initial design. They represent a snapshot in time.

Solution:

In the physical reality of industrial operation, safety integrity is not a static attribute. It is a dynamic state. It is continuously reshaped by the cumulative effects of operational demand, environmental stressors, and specialized maintenance intervals. This framework introduces a theory of Temporal Safety Integrity. At its core is the concept of Mosaic Aging.

Market Comparison Matrix

Feature	Competitors	SIL10
Update Frequency	Annual	Real-time (30 sec)
SIL Vision	Static "SIL 2"	Dynamic SIL(t) trajectory
Degradation Visibility	None	5 factors shown separately
Common-Cause Tracking	Assumed $\beta=0.10$	Dynamic $\beta(t)$ evolution
Diagnostic Coverage	Constant	Drifting DC(t) tracked
Forecasting	None	36-month prediction
Uncertainty Quantification	None	100-scenario Monte Carlo
Action Recommendations	None	Ranked by ROI

Market Comparison Matrix

Feature	Competitors	SIL10
Audit Trail	Paper-based	Immutable SHA-256 hash chain
Compliance Proof	Annual audit	24/7 continuous monitoring
IEC 61508 Mapping	Manual verification	Automatic per calculation
Recovery Capability	Manual/untested	Tested, verified, automated
Regulatory Defense	"We assessed it"	"14.2 M proof entries available"
Cost-Benefit Analysis	Qualitative	Quantitative (ROI calculation)
Production Integration	Export/manual	Native OPC-UA, MQTT, REST APIs

Motivation of SIL → SIL(t)

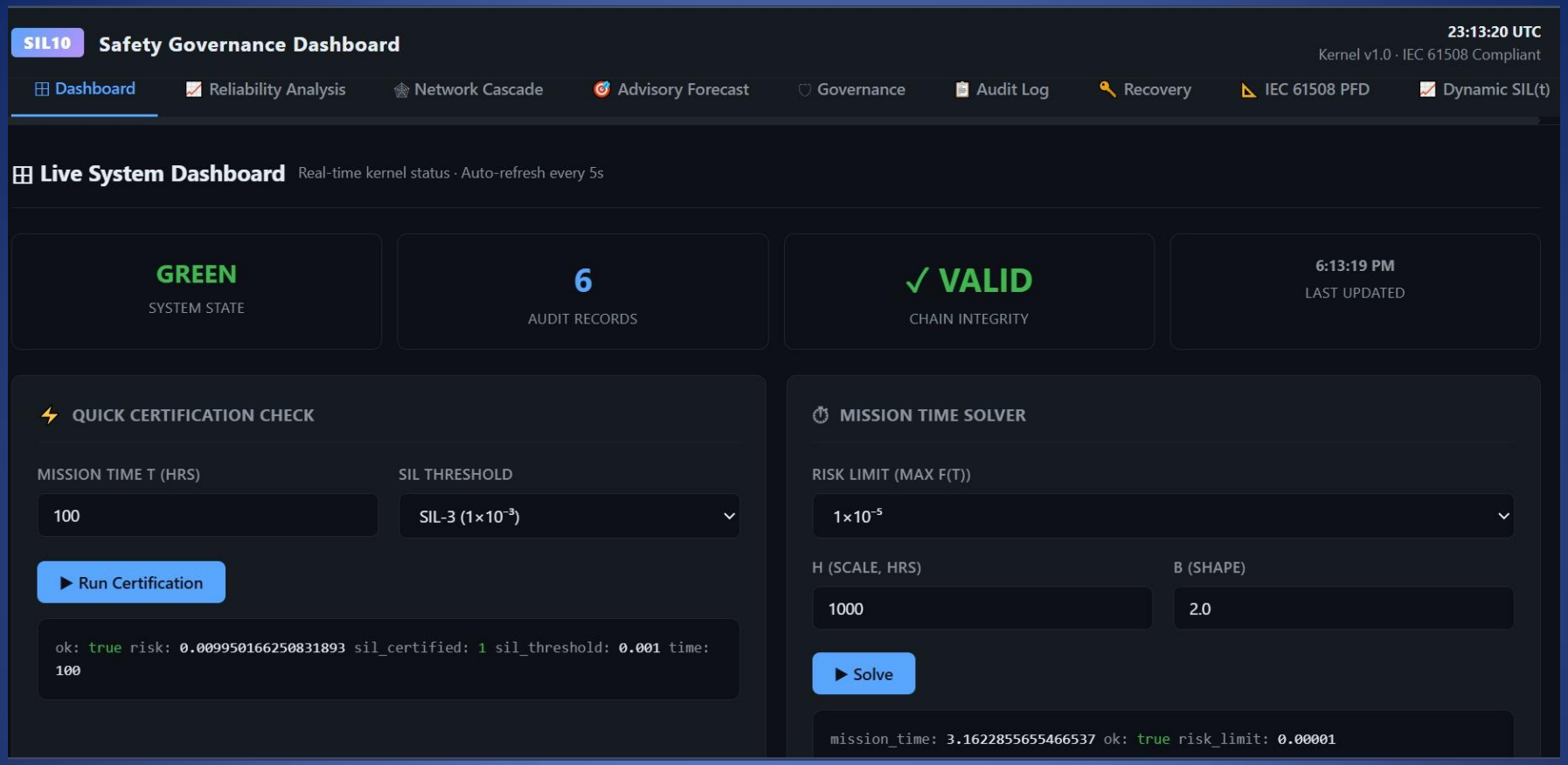
Consider SIL10 now, or wait for an incident to force a 360-degree realization. One path is proactive and other is painful.

It Measured in:

Loss of life,

Loss of money,

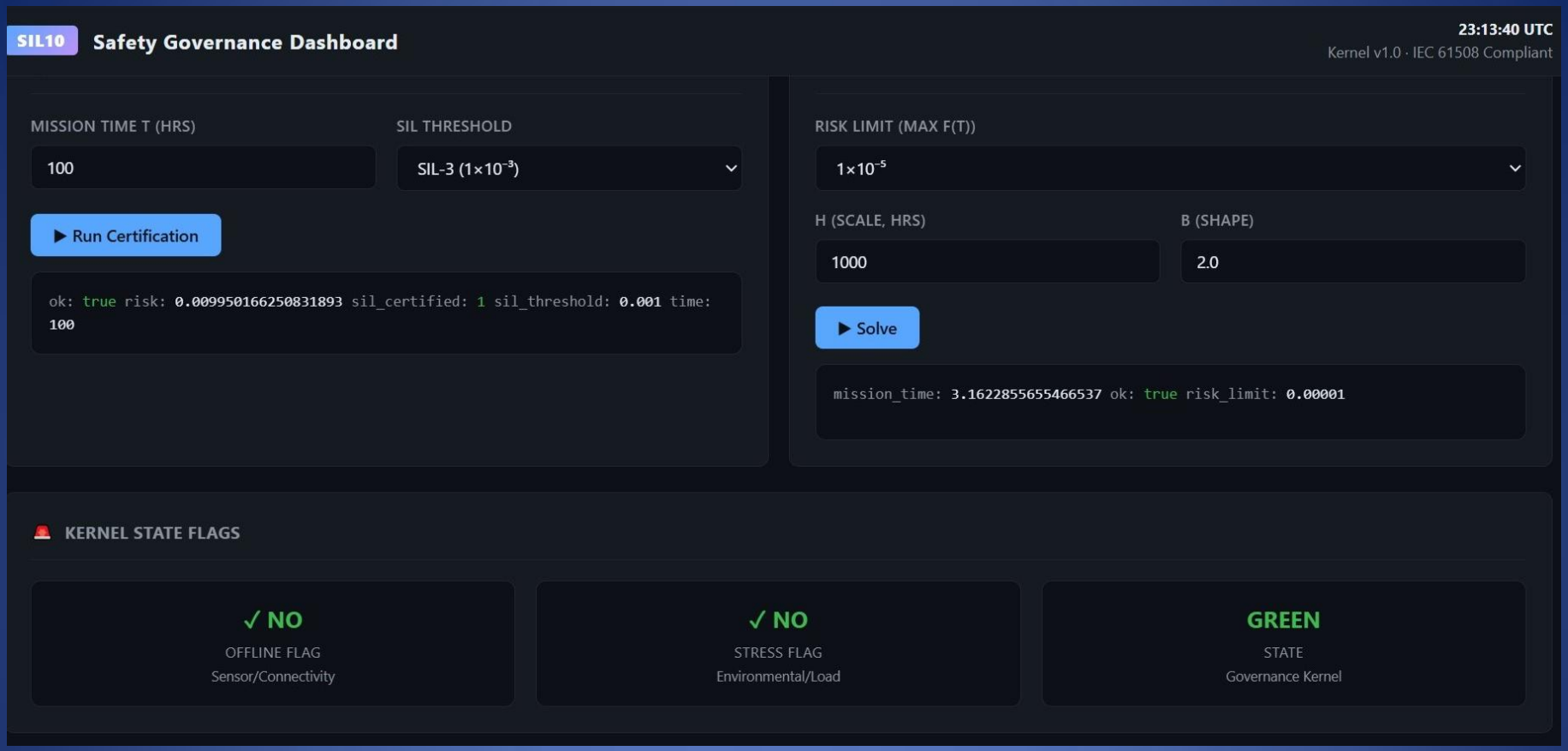
Adverse consequences no plant can afford.



Tab 1 = Main Dash Board 1/2

Plant-wide safety health at a glance
Shows: Risk score (%), loop count by SIL, alert feed, 90-day trend
Graphs: Risk timeline, SIL distribution (donut), alert feed
Live Connection: Real-time, <2 sec latency (Kafka Data Engineering)

by Usman Zafar PhD



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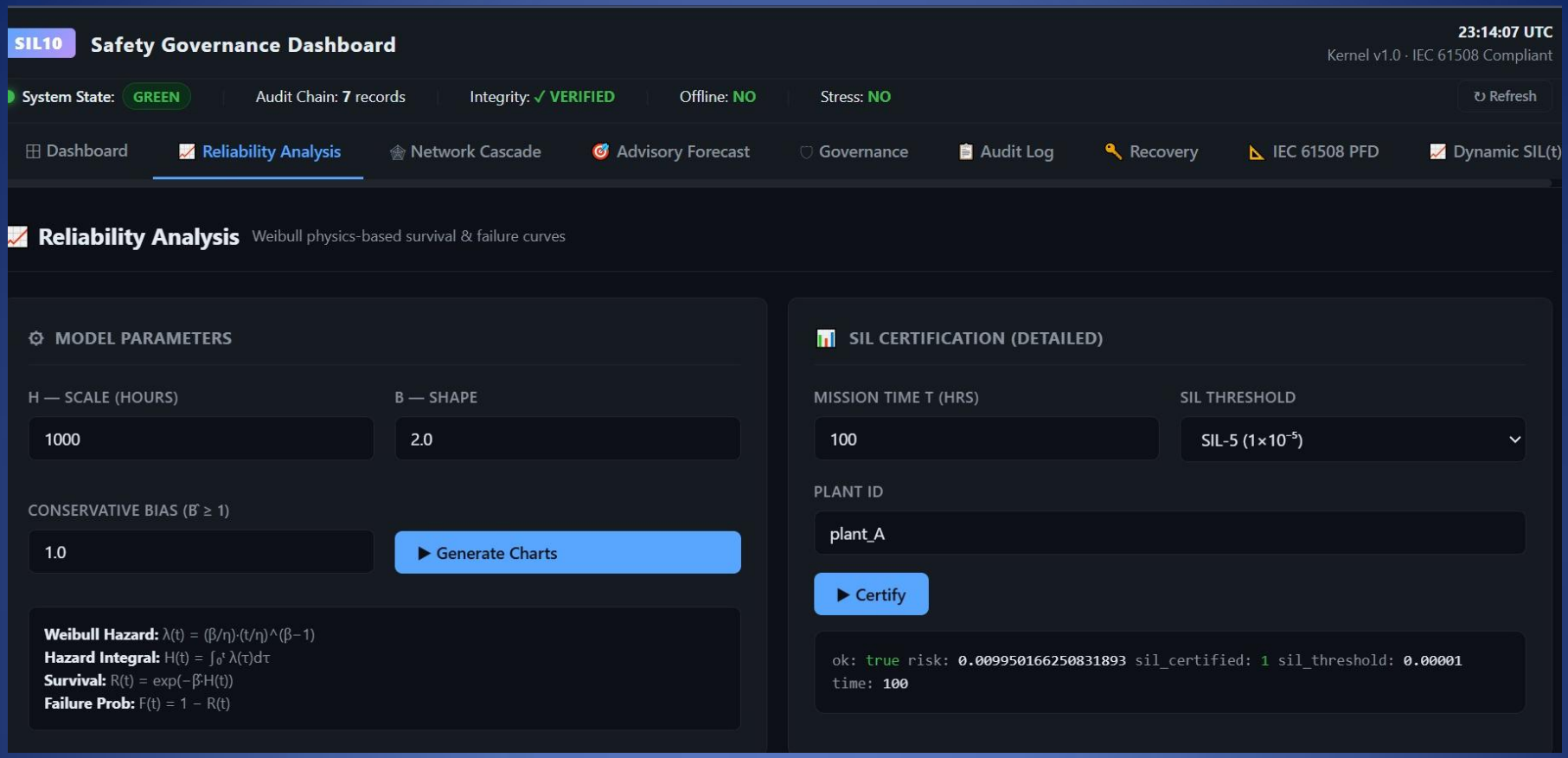

Tab 2 = Reliability Analysis 1/2

What: Why is SIL degrading? Detailed component breakdown
Shows: λ_D, DC, β, 5 degradation factors per channel, 1oo2 effectiveness
Graphs: Hazard rate evolution, factor stacking chart, DC drift, β(t) coupling

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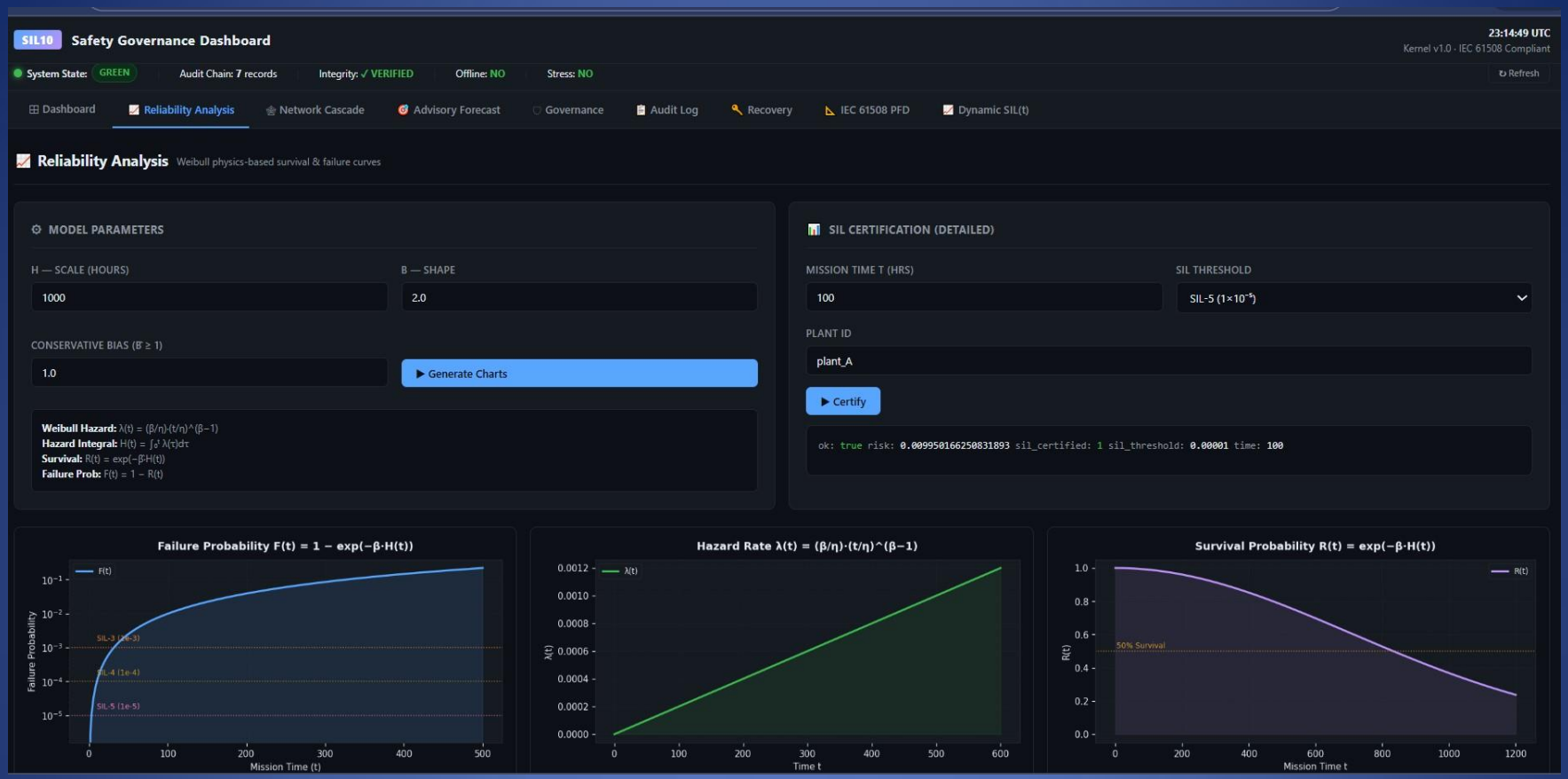

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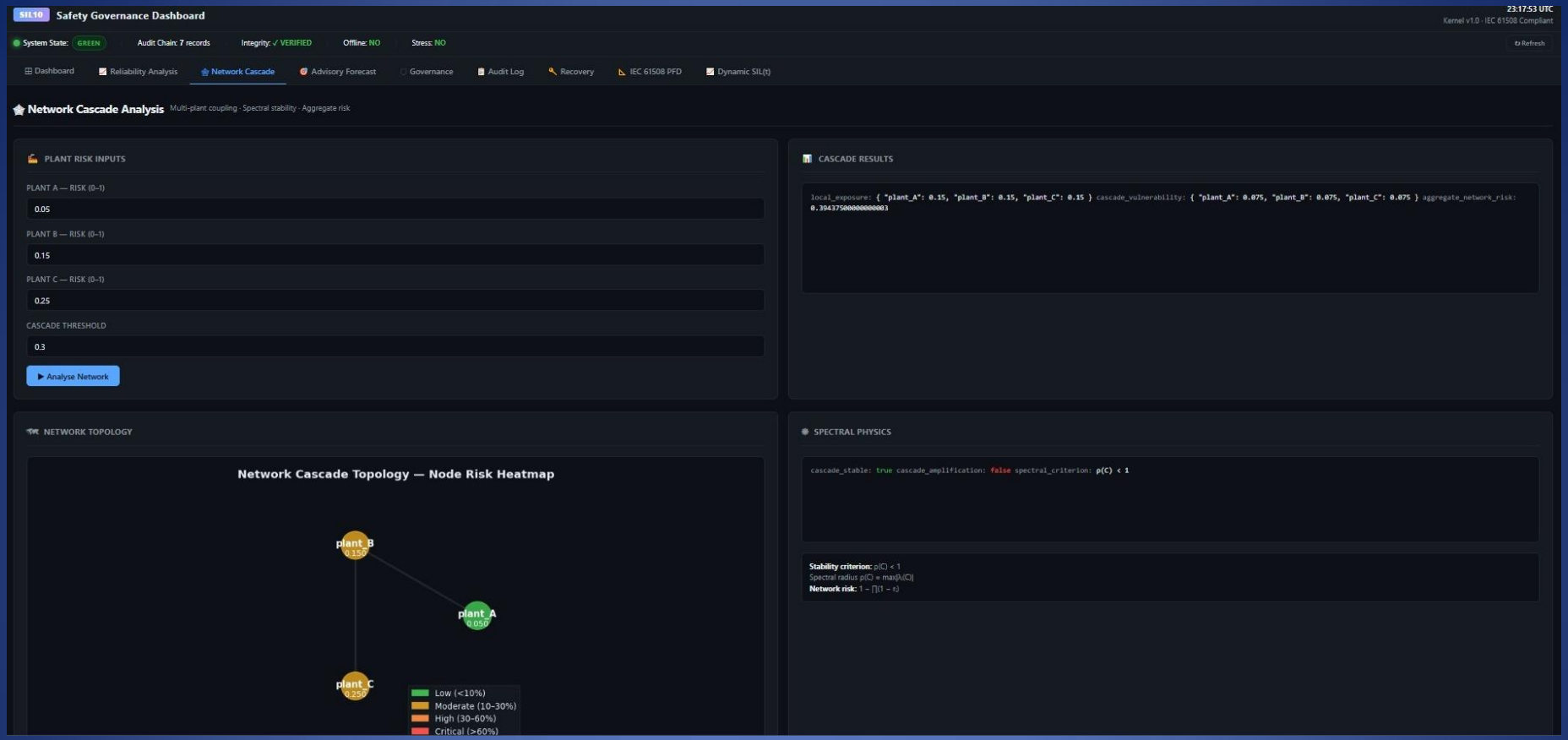

TAB 3: NETWORK CASCADE

What: If component X fails, what else dies?
Shows: Dependency graph (component → loop → plant), cascade impact (SIL drops), vulnerability score
Graphs: Network topology map, failure impact analysis, timeline to criticality

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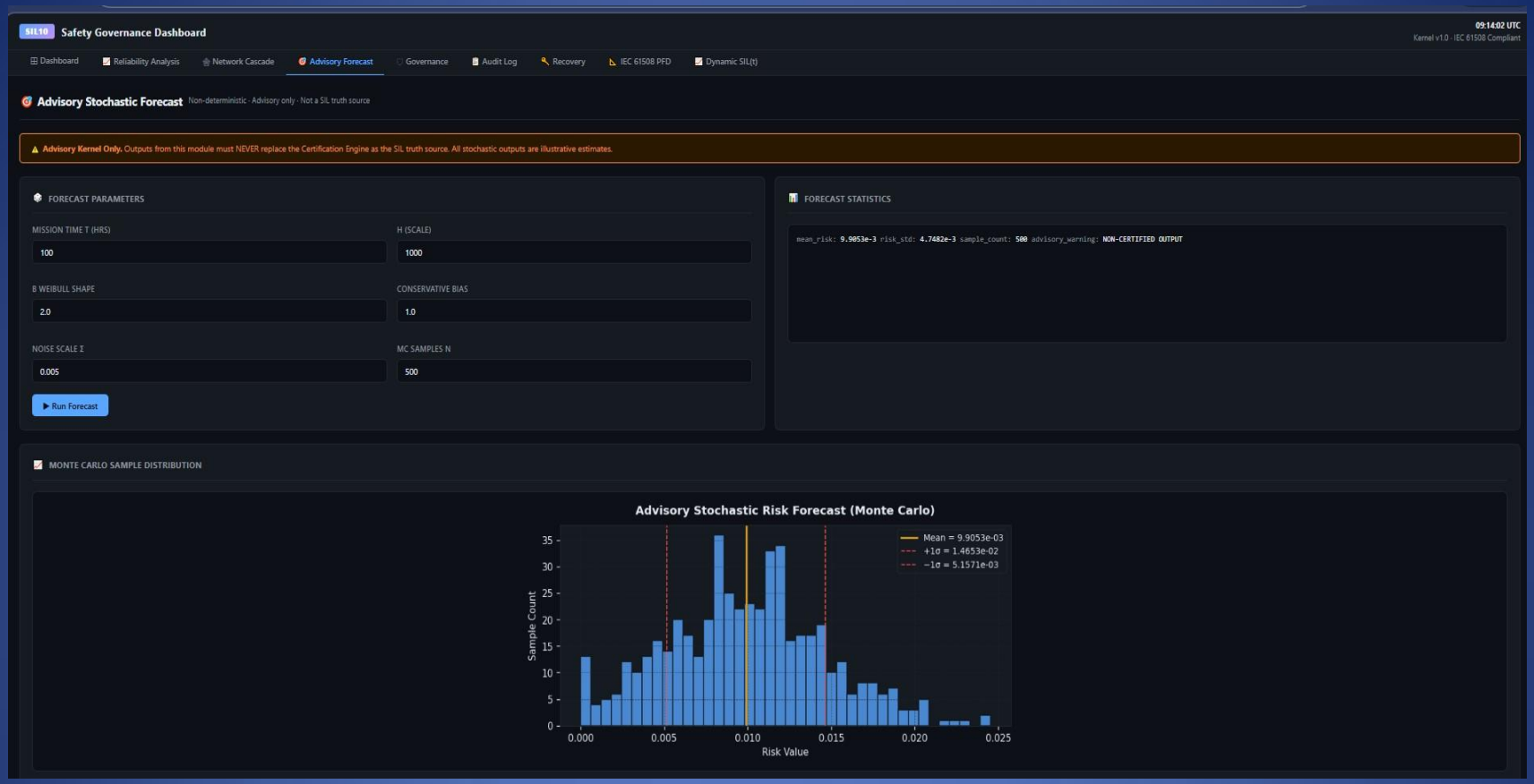

TAB 4: ADVISORY FORECAST

What: What will SIL be in 36 months? What should you do?
Shows: SIL trajectory, uncertainty bands (5th-95th %ile), action plan with cost/ROI
Graphs: SIL forecast curve, Monte Carlo distribution (100 scenarios), recommendation panel, intervention impact

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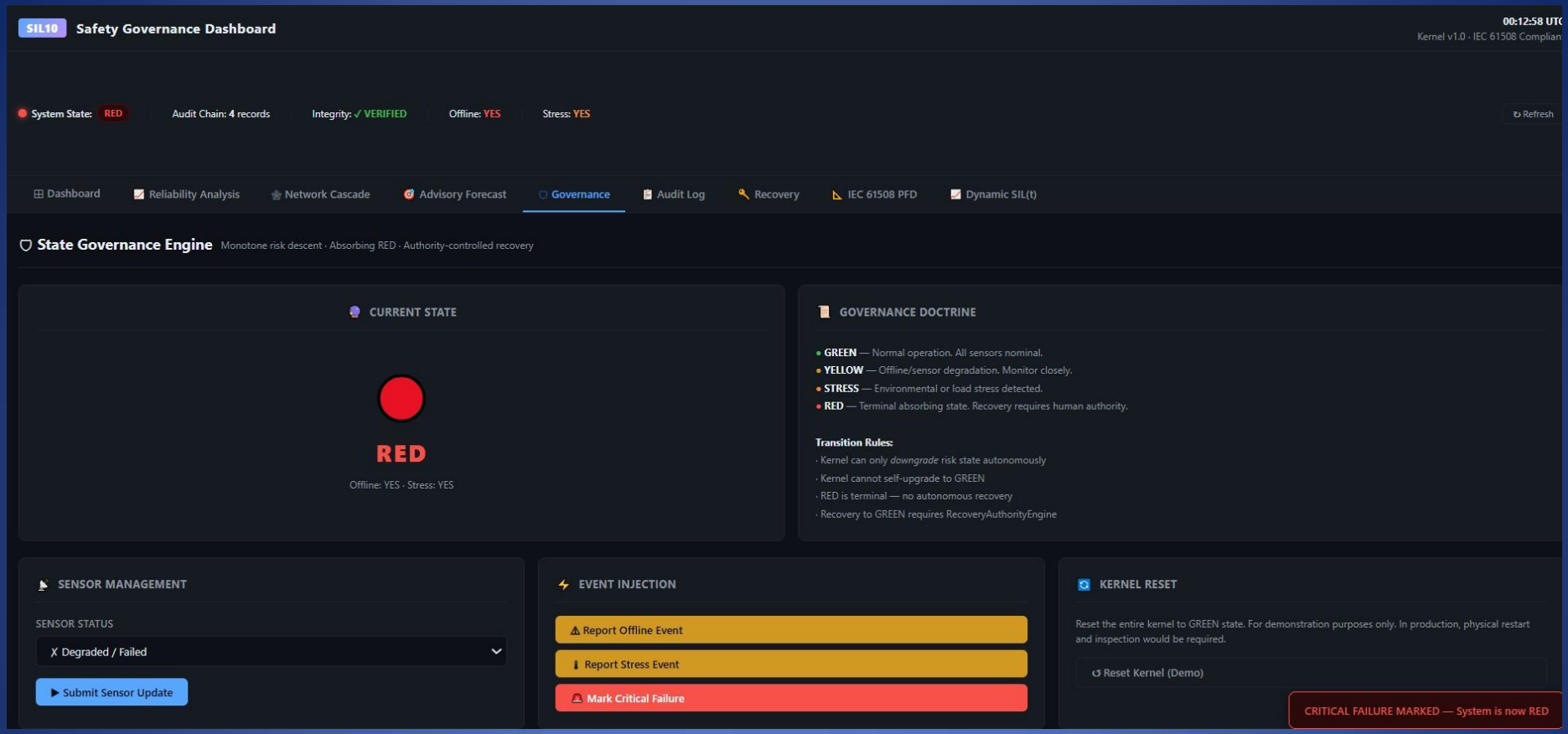

TAB 5: GOVERNANCE

What: Prove to auditors you're monitoring continuously
Shows: Compliance status, monitoring duration, calculation count, alert/action logs, SIL maintenance score
Graphs: Compliance timeline, governance status panel, metrics dashboard

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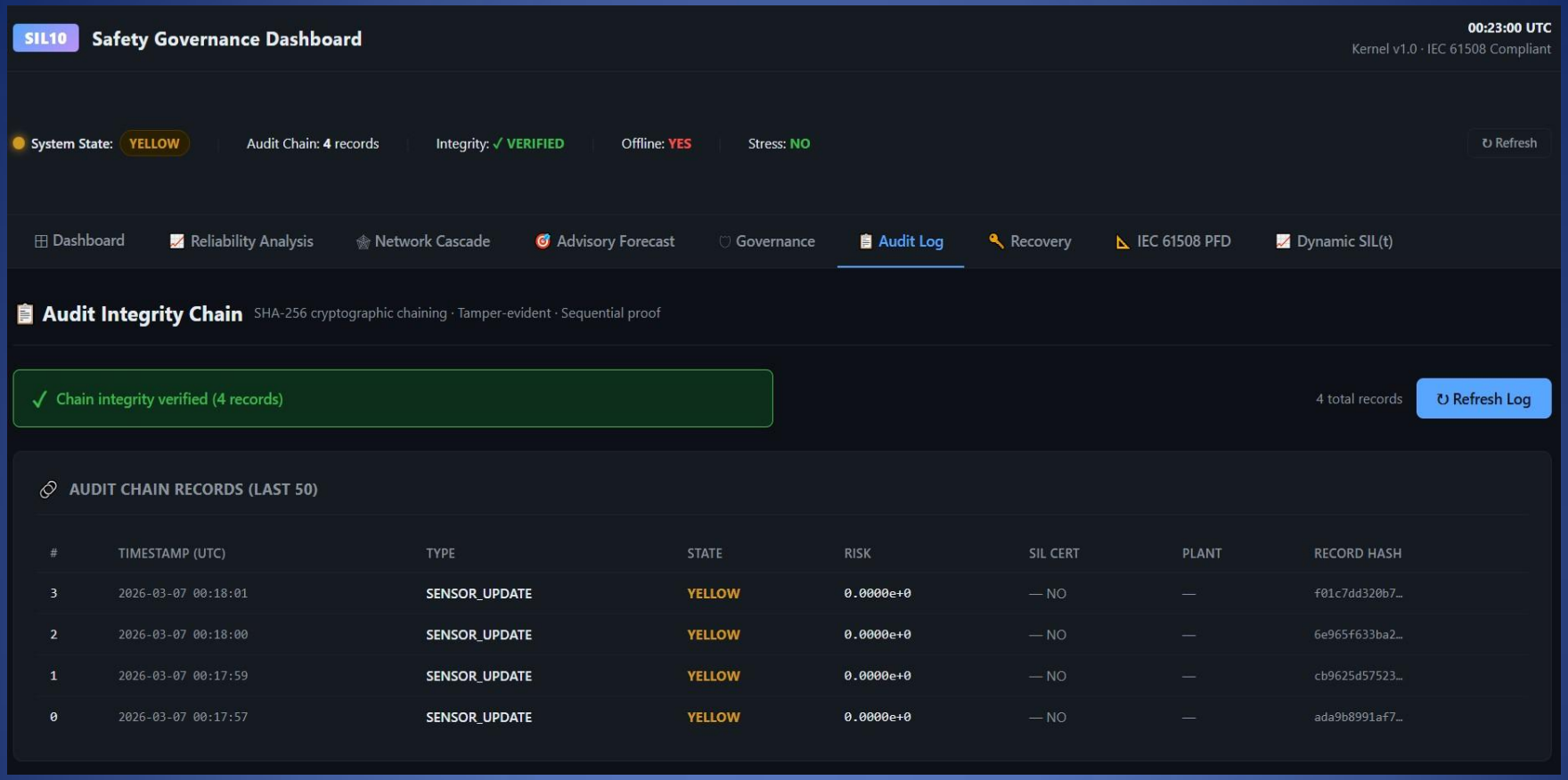

TAB 6: AUDIT LOG

What: Immutable record of every decision (SHA-256 hashed)
Shows: Complete calculation history, input state, output state, change from previous, hash chain verification
Graphs: Timeline browser, entry detail view, data integrity verification

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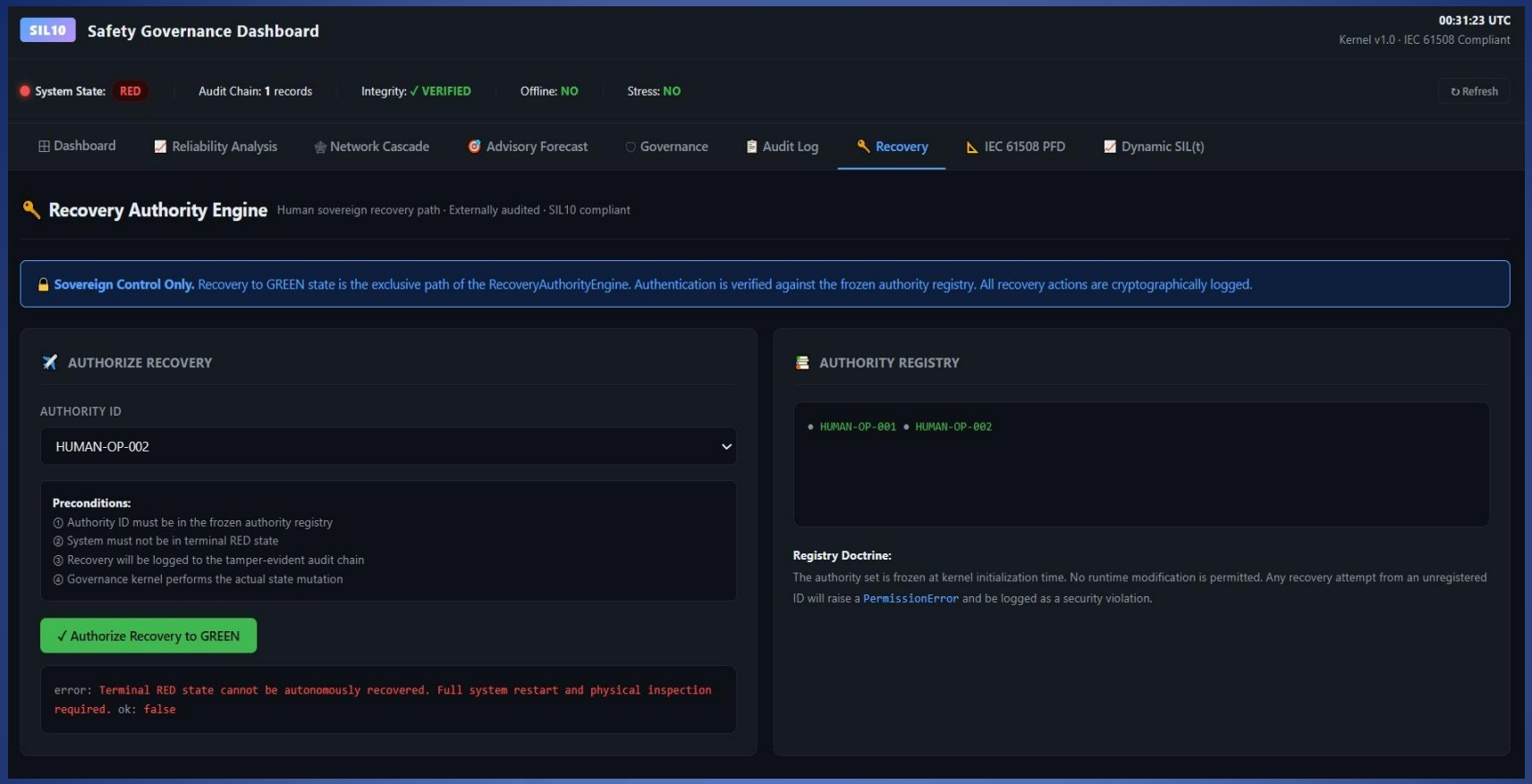

TAB 7: RECOVERY

What: System crashed? How do we restore?
Shows: Backup status (PostgreSQL, InfluxDB, Kafka), RTO/RPO, recovery procedures, integrity verification
Graphs: Backup timeline, restoration procedure flow, Kafka offset tracking

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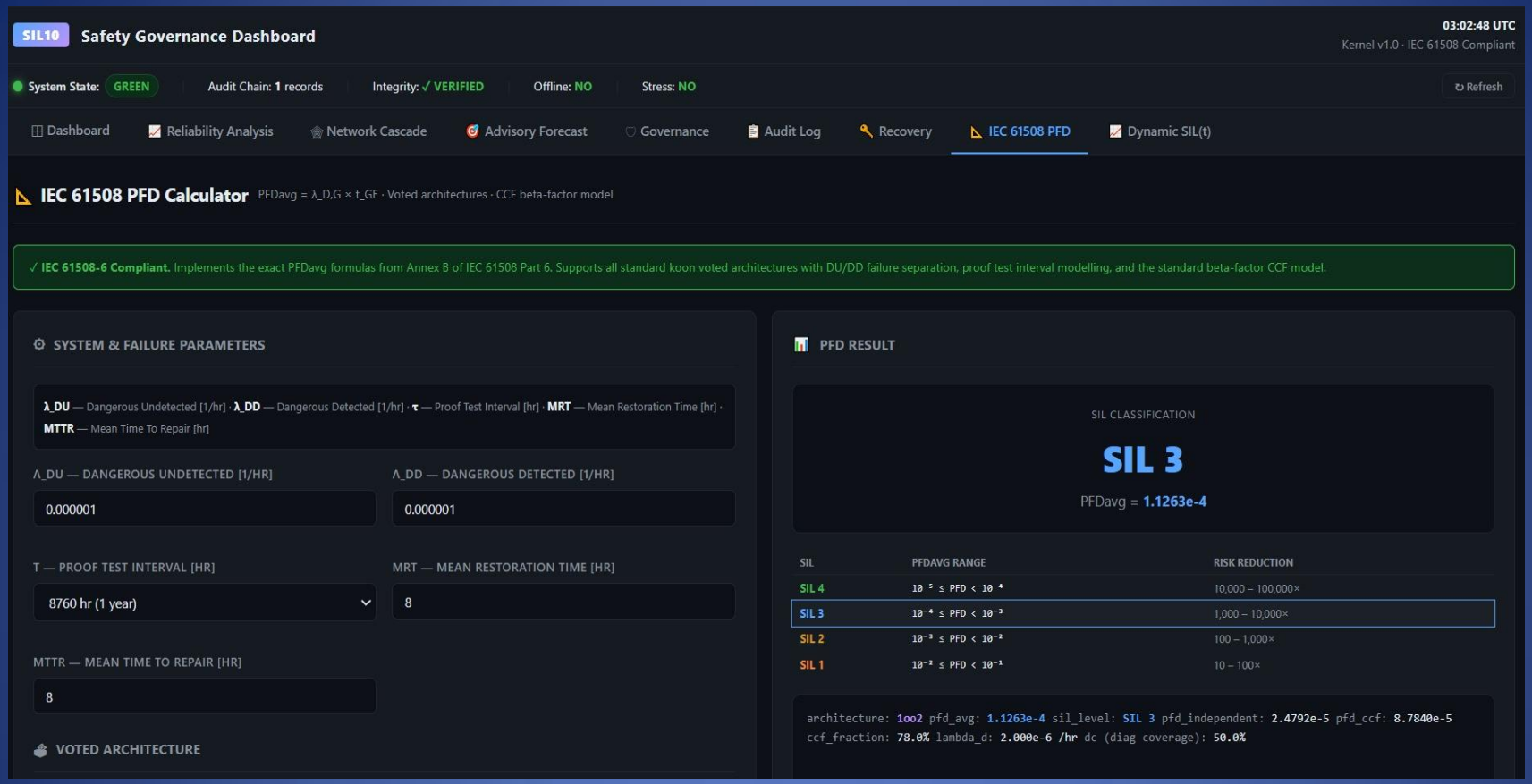

TAB 8: IEC 61508 PFD 1/3

What: Translate our calculations to official IEC 61508 terminology
Shows: Formula mapping, step-by-step calculation per IEC standard, SIL classification audit certificate
Graphs: Formula reference display, compliance checklist, architecture-specific calculation flow

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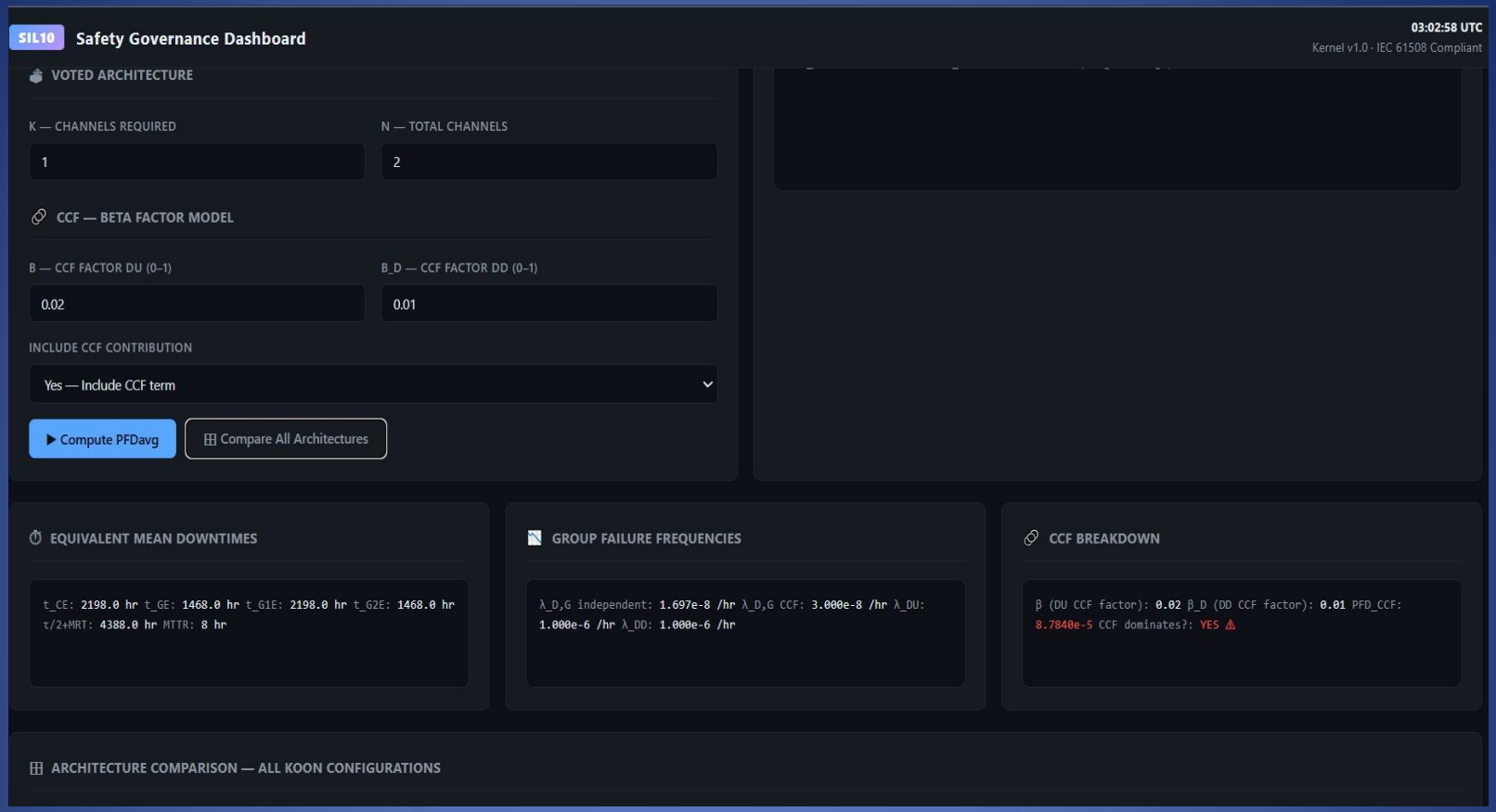

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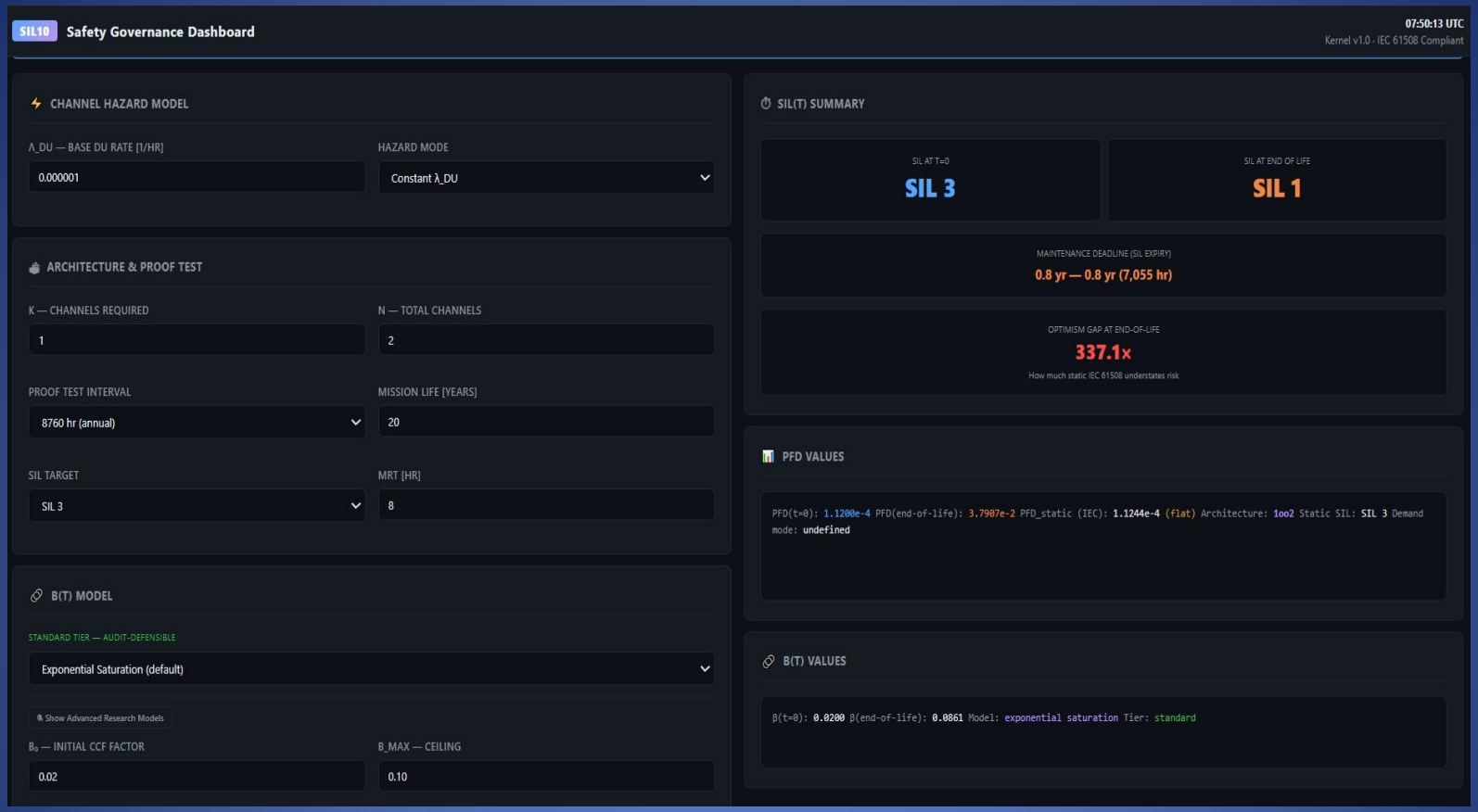

TAB 9: DYNAMIC SIL(t)

What: Not "what is SIL today" but "what will SIL be" → strategic vision for 36 months
Shows: SIL trajectory with decision gates, uncertainty bands, critical thresholds marked, year-by-year snapshots
Graphs: Main SIL(t) trajectory, Monte Carlo distribution, comparison (today vs year 1 vs year 5), intervention impact

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